

## Laboratory and Field Test of a Sampling Method for Characterization of Combustion Aerosols at High Temperatures

M. Strand,<sup>1</sup> M. Bohgard,<sup>2</sup> E. Swietlicki,<sup>3</sup> A. Gharibi,<sup>3</sup> and M. Sanati<sup>1</sup>

<sup>1</sup>*Division of Bioenergy, Växjö University, Växjö, Sweden*

<sup>2</sup>*Division of Ergonomics and Aerosol Technology, Lund University, Lund, Sweden*

<sup>3</sup>*Division of Nuclear Physics, Lund University, Lund, Sweden*

The objective of this study was to design and experimentally examine a sampling method for high-temperature aerosols from biomass combustion, in which nucleation and condensation from fly ash forming vapors is controlled. The sampling method includes a high-temperature probe in which the hot gas is diluted and then cooled. Laboratory results from sampling a model aerosol with known concentrations of SiO<sub>2</sub> particles and KCl vapor showed that when using a high dilution ratio, the KCl vapor was effectively separated from the aerosol by deposition onto the probe walls. When a lower dilution ratio was used, the KCl vapor generated a distinct nucleation mode when cooled in the probe. The sampling method was also used for sampling flue gas from a circulating fluidized bed boiler fired with forest residues. The results suggest that the major fractions of Ca, K, S, and Zn were present as particles at 780°C, whereas most of the Cl and Pb were present as gases. The field results are consistent with results previously reported and indicate that the method can be used for efficient and precise characterization of high-temperature combustion aerosols containing inorganic vapors.

### INTRODUCTION

Combustion of biomass for heat and power production in medium-scale-to-large-scale units is today a well-established and growing activity in several regions of the world. Potential solid biomass fuels, such as wood and various agricultural by-products, usually contain a substantial fraction of ash-forming inorganic compounds that might be detrimental to process and plant, causing operational problems such as corrosion and blockage (Miles et al. 1996; Baxter et al. 1998; Michelsen et al. 1998).

Some of the material will be transported through the boiler, subsequently being released as particulate emissions to the ambient atmosphere. The submicron fraction of the particulate emissions from combustion processes have a long residence time in the atmosphere and might be associated with adverse health effects due to their penetration and deposition in the lower respiratory tract, and their enrichment of potentially toxic components (Lighty et al. 2000).

Large-scale combustion of wood has been shown to produce a fly-ash mass size distribution with two distinct modes. (Hasler and Nussbaumer 1998; Valmari et al. 1998; Obernberger et al. 2001; Strand et al. 2002; Pagels et al. 2003). The particles in the two modes have different chemical compositions due to separate mechanisms of formation. The fine-mode particles in the submicron size range ( $<1\ \mu\text{m}$ ) are formed mainly from constituents that are vaporized during combustion of the fuel, e.g., K, S, Cl, and heavy metals. The vaporized components are subject to homogenous chemical reaction and might form vapors with low saturation vapor pressure, leading to gas-to-particle conversion at high temperatures. During combustion of straw, the main constituents of the submicron particle fraction were potassium sulphate and chloride (Christensen et al. 1998). A theoretical analysis indicated that the gas-to-particle conversion occurs during the cooling of the flue gas by homogeneous nucleation of K<sub>2</sub>SO<sub>4</sub> particles, which act as condensation nuclei for the subsequent condensation of KCl. The supermicron particles ( $>1\ \mu\text{m}$ ) are formed mainly from ash particles entrained from the fuel bed, and they consequently have a higher concentration of refractory ash components such as Ca, P, and Mn.

Pulverized coal combustion generally generates a fine-particle mode around  $0.1\ \mu\text{m}$ . Coal contains only low concentrations of volatile alkali compounds, but in the high temperature and reducing atmosphere surrounding the char particle the refractory oxides (e.g., Si, Al, Fe, and Mg) can undergo

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Address correspondence to Michael Strand, Växjö University, IBP, Department of Bioenergy Technology, SE-351 95 Växjö, Sweden. E-mail: michael.strand@ibp.vxu.se

a reduction to more volatile suboxides or metal compounds that might vaporize (Flagan and Seinfeld 1988). It is not fully investigated if a fine-particle mode can be formed also during biomass combustion by reduction of refractory oxides close to the burning char. When firing bark in a moving-grate, 450 KW boiler, considerable amounts of Ca were found in the submicron mode (Obernberger et al. 2001). For other fuels the concentration of Ca was much lower. The authors assumed that a release of Ca to the vapor phase was not possible and that the Ca-containing submicron particles were released directly from the fuel.

In order to achieve a deeper understanding of the particle formation process in biomass combustion, it is desirable to have a method for characterization of combustion aerosols at high temperatures, where a fraction of the ash-forming material is still present as inorganic vapors. There are certain properties that must be considered when sampling and measuring aerosols of high concentration and high temperature, since a number of dynamic physicochemical processes are in progress in the aerosol system. If a sampling and cooling device is used, it must be designed to quench all aerosol dynamics and chemistry in order to obtain a representative measurement. Quenching will constitute a specific problem when the combustion aerosol contains inorganic vapors, e.g., alkali or heavy metal vapors. These vapors generally have a negligible vapor pressure at the temperature prevailing at the probe outlet. Consequently, they will condense as the gas temperature is decreased in the probe, which might affect the original particle phase. The condensing vapors might contribute to the particle phase by deposition on the surface of the original particles and by forming a nucleation mode. In both cases, the use of the sampling device will alter the original size distribution and particle composition.

Recently there has been increasing interest in dilution effects from internal combustion engines, especially diesel engines (Kittelson 1998). When sampling the exhaust aerosols, temperature, concentration, and humidity must generally be decreased upstream from the instruments. When diluted, condensing species, e.g., volatile organic and sulphuric compounds, might form high number concentrations of ultrafine particles. The present interest in the dilution effects is not only a matter of conserving the particle concentration in the hot exhaust gas but also of simulating the aerosol dynamics at the tailpipe outlet. In one study, a two-stage diluter was used when sampling the exhaust from a diesel engine (Kittelson and Abdul-Khalek 1999). The primary diluter was designed to produce high saturation ratios by diluting with cool gas. The second diluter was used in order to quench the aerosol dynamics. The results showed that the formation of ultrafine particles depended on dilution ratio and temperature as well as concentration of particles and particle precursors.

Diffusion denuders have been used frequently in order to separate condensable gas components prior to the cooling of the aerosol (Willeke et al. 1993). Traditionally designed diffusion denuders are not suitable for high-temperature ( $>600^{\circ}\text{C}$ )

biomass combustion applications, since the temperature as well as the complexity of the gaseous components makes it difficult to design appropriate adsorption coatings.

There have been a few attempts to characterize high-temperature aerosols from biomass combustion. Valmari et al. (1998) studied fly-ash particles formation and fly-ash deposition during fluidized bed (CFB) combustion of willow at  $650^{\circ}\text{C}$ . The mass size distribution was measured using an in-duct, 11-stage, multijet, Berner-type, low-pressure impactor. The author's interpretation of the results was that the mass size distribution included inaccuracies since the collection substrates gained mass due to reactions with components of the hot flue gas.

In another study of ash behavior during CFB combustion of biomass, Valami et al. (1999a) investigated the gas- and particle-phase concentration of ash-forming compounds at  $800\text{--}850^{\circ}\text{C}$ . The particles were collected in an in-duct sampling system consisting of a precyclone and a quartz filter. After the quartz filter, the flue gas sample was quenched with cool air using a porous tube in order to minimize sampling losses. During quenching, the vapors formed fine particles that were collected on a polycarbonate filter. The elemental composition of the gas phase was determined by analyzing the particles collected on the second filter. The deposited particles were composed mainly of K and Cl. Sulphur was not found in significant amounts, indicating that the sulphur was present mainly as particles collected on the first filter or as gases ( $\text{SO}_2$ ) not condensed during quenching. According to the authors the vapor fraction might be underestimated because some of the vapors might have reacted with the first quartz filter and been erroneously counted as particles.

In order to understand the behavior of fly-ash-forming constituents at high temperatures, it is desirable to have a simple method of investigating how the fly-ash-forming compounds are distributed between vapor phase and particles phase, as well as between particles of different size fractions. The method should allow for efficient and reliable vapor-particle separation at positions close to the burning fuel, where the ratio of vapor to particle concentration is expected to be high. Moreover, the method should allow for transient measurements using on-line instruments, as enhanced variations in concentration and size distribution can be expected at positions of early particle formation.

The objective of this study was to investigate experimentally the possibility of using a high-temperature dilution-deposition probe in order to control the separation the fly-ash-forming vapors from flyash particles at high temperatures. The method was examined using a laboratory setup where a well-defined model aerosol was generated and sampled. The sampling technique was also tested in the field, characterizing particles and vapors at  $780^{\circ}\text{C}$  in a 104 MW CFB boiler fired with wood chips. The results show that using a high-temperature dilution probe is a feasible method for sampling aerosols in biomass-fired boilers, enabling online characterization at high temperatures.

## EXPERIMENTAL

### *The High-Temperature Dilution Probe*

The sampling probe consists of two coaxial quartz tubes (Figure 1). The length of the outer tube is 0.8 m, with an inner diameter (i.d.) of 28 mm. At the hot side of the outer tube, there is a contraction ending with a 25 mm long, 5 mm i.d. inlet tube. At the cold end of the 28 mm tube there is a contraction in order to seal the outer tube around the inner tube using a rubber ring sealing. At the cold end there is also a perpendicular 8 mm inlet tube for the dilution gas. The inner tube (i.d. 9 mm) is centered inside the outer tube by a sleeve. The inner tube can be slid axially, varying the width of the circular slit that is formed between the tip of the inner tube and the hot side contraction of the outer tube. During sampling the slit width was kept below 1 mm in order to accomplish proper mixing conditions in the mixing zone. The dilution air was fed through the perpendicular inlet and heated by tube walls as it was transported towards the dilution position at the hot end of the probe. In the dilution and mixing section the gas from the inlet tube was diluted with the heated dilution air. The diluted gas was then cooled to about 200°C by the walls before exiting the probe. The gas was further cooled to 25°C before being split between the instruments using a copper tube cooled in a water bath. Dried and filtered compressed air was used as dilution gas, and the flow rate was controlled using a calibrated mass flow controller

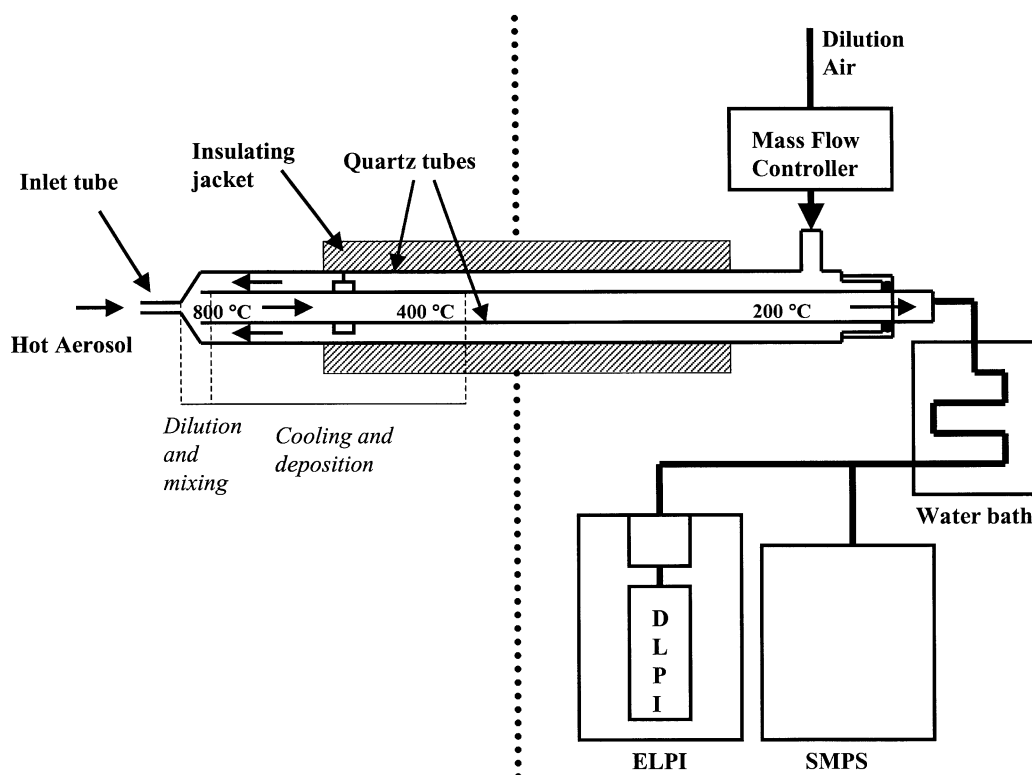
(Bronkhorst HI-TEC). The outlet flow to the instruments was 10 l/min.

In order to control the dilution ratio, the mass flow controller was first set to 11 l/min, which produced a net flow of dilution air out of the probe tip. By decreasing the rate of dilution air stepwise until the instruments started to indicate particles, the point of zero-flow through the inlet tube could be established. By further decreasing the flow rate of dilution air, the flow rate of undiluted gas through the inlet tube could be controlled, and the desired dilution ratio could be set. The procedure of establishing the zero-flow was necessary in order to produce high dilution ratios, since the flows through the mass flow controller and the instruments might vary slightly due to drift and changes in ambient pressure and temperature. Nevertheless, by initiating each measurement with establishing the zero-flow, a stable flow difference as low as 0.1 l/min could be produced, yielding a dilution ratio of 1:100 with an accuracy of 30%.

When sampling from the CFB boiler, an additional insulating jacket was used. The jacket could be slid along the probe in order to obtain an appropriate temperature profile in the probe, even when a large part of the probe was exposed to the hot flue gas.

### *Particle Characterization*

An electrical mobility spectrometer (SMPS; 3934, TSI Inc.) incorporating a condensation particle counter (CPC;



**Figure 1.** The sampling system used for high-temperature aerosol characterization. The aerosol was diluted and cooled inside the probe. Particle samples were collected using the DLPI, and the number size distributions were measured using the SMPS system.

model 3022, TSI Inc.) was used for determining the particle number size distribution. The size interval studied was in the range of 17–550 nm. In addition to the SMPS, an electric low-pressure impactor (ELPI; Dekati Ltd.) was used. The ELPI system used gives a size distribution in the 0.03–10  $\mu\text{m}$  range in 12 channels. Because of the low channel resolution in the 30–100 nm range, the ELPI was not suitable for accurately determining the size distribution of the nucleation mode. However, the ELPI has a high time resolution and a constant flow of about 10 l/min, and the ELPI instrument was therefore used for establishing the zero flow and the dilution ratio as well as for providing a constant flow through the dilution probe.

In order to obtain size-classified particle samples from the CFB boiler for elemental and gravimetric analysis, the internal ELPI Dekati low-pressure multijet cascade impactor (DLPI) was used. The DLPI classifies the aerosol particle into 13 size fractions ( $d_{50}$  0.03–10  $\mu\text{m}$ ). The sampling was carried out by first establishing the desired dilution ratio, operating the ELPI in the ordinary online sampling mode. Then the internal corona charger and the ion trap were turned off, allowing the ELPI to be used as an ordinary DLPI. The high-dilution samples and the low-dilution samples were collected subsequently within a period of 4 h. Only the first 7 stages of the impactor, corresponding to the submicron particle fraction, were analyzed. Greased polycarbonate foils were used as impactor substrates. Particle-induced X-ray emission analysis (PIXE) was used for elemental analysis of the impactor substrates. Approximately 30 elements were analyzed. Since PIXE is not suitable for elements with an atomic number less than 13, ash components such as Na and Mg were not analyzed.

### The Laboratory Setup

A model aerosol with  $\text{SiO}_2$  particles and KCl vapor was generated in the laboratory setup (Figure 2). The  $\text{SiO}_2$  particles were generated by decomposition of Hexamethyldisiloxane at 650°C in tubular furnace (reactor 1). The atomizer was used to generate a spray of KCl dissolved in water. The KCl solution droplets were mixed with the aerosol from reactor 1 before being

fed into the second tubular furnace reactor (reactor 2). In reactor 2 the KCl solution droplets were dried and then evaporated at 900°C. The aerosol with  $\text{SiO}_2$  particles and KCl vapor was sampled inside reactor 2 at a temperature of 900°C, using the high-temperature dilution probe. The dilution ratio in the probe was varied in order to investigate the dilution effect. Measurements were carried out when generating  $\text{SiO}_2$  particles or KCl vapor only, as well as when generating both particles and KCl vapor.

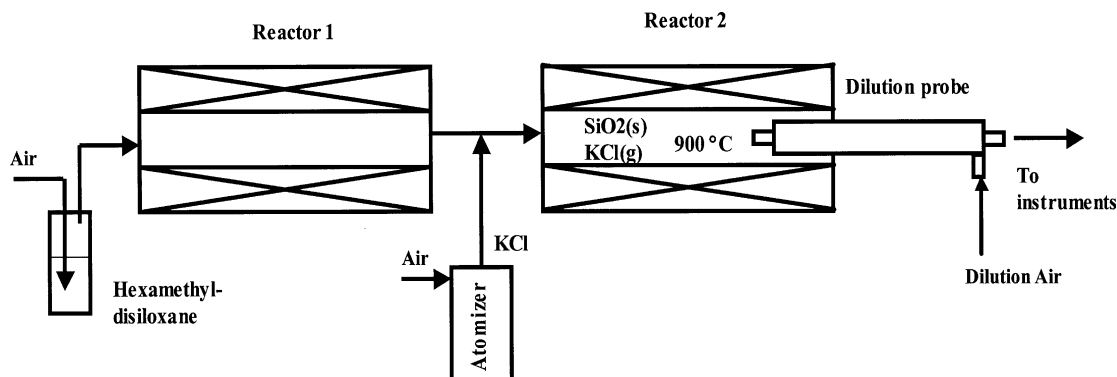
### The CFB Boiler Setup

The flue gas was sampled in a boiler fired with forest residues mixed with about 10% peat. The boiler is a circulating fluidized bed type with a thermal effect of 104 MW. The flue gas sample was extracted from the duct between the process cyclone and the super-heater at a temperature of about 780°C. The sample was taken perpendicular to the direction of the flue gas flow in order to avoid overloading and clogging by coarse fly-ash particles and bed material. Only the particle fraction <1  $\mu\text{m}$  was studied. The inlet efficiency of the probe was estimated to be close to 1 for particles smaller than 0.5  $\mu\text{m}$ .

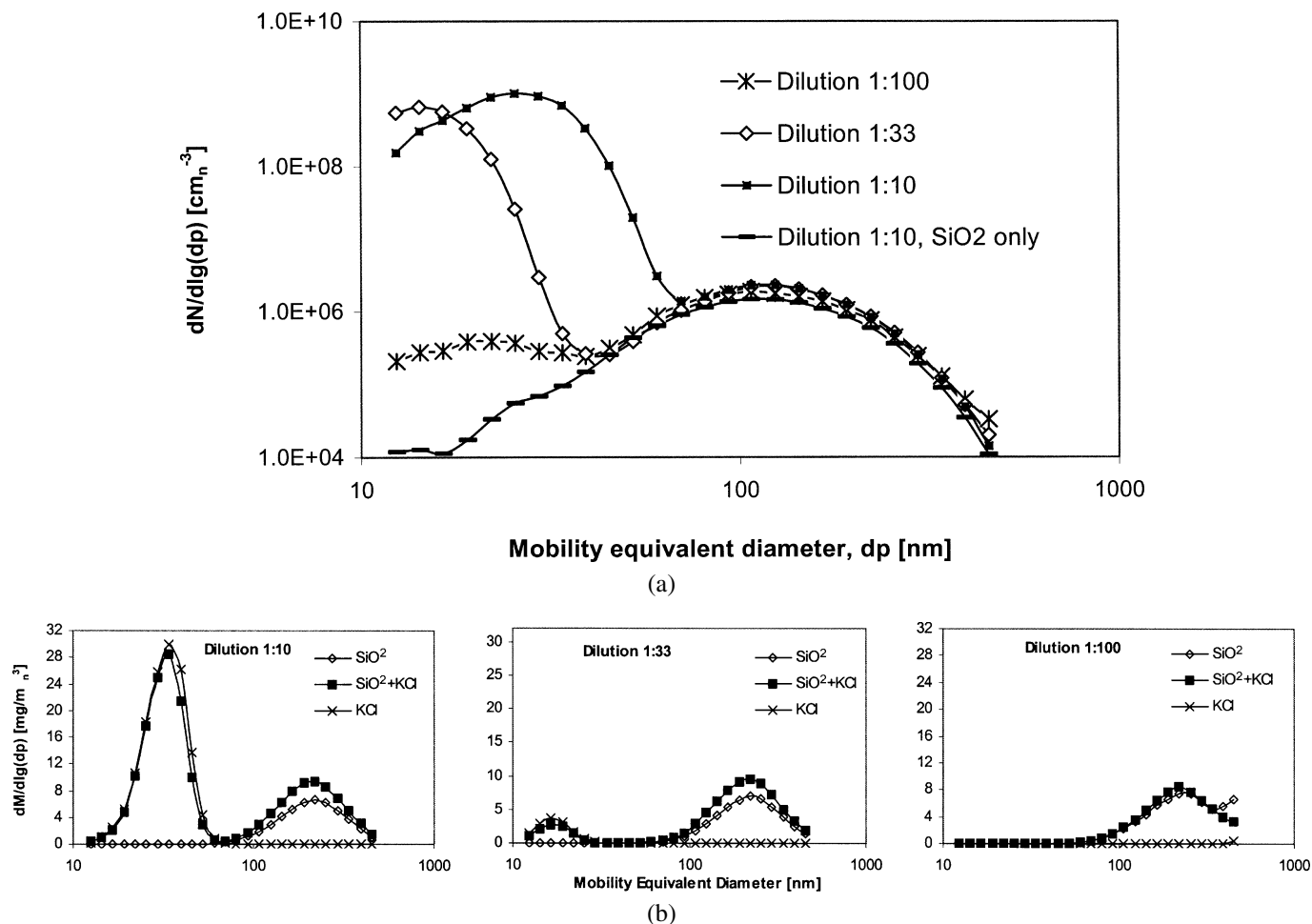
## RESULTS AND DISCUSSION

### Results from Laboratory Setup

The number size distributions obtained for different dilution ratios are given in Figure 3a. All size distributions given in all figures are corrected for dilution. When feeding  $\text{SiO}_2$  particles only, and using a dilution ratio of 1:10, the number size distribution was unimodal, with a geometric mean diameter (GMD) of 110 nm and a number concentration of  $4 \times 10^5$  particles/ $\text{cm}_n^3$  ( $1 \text{ cm}_n^3 = 1 \text{ cm}^3$  at 0°C, 1 atm). When adding 20 mg/ $\text{m}_n^3$  of KCl vapor to the reactor, and using the same dilution ratio (1:10), a distinct nucleation mode around 30 nm was generated in addition to the original mode. As the dilution ratio was increased to 1:33 when feeding both  $\text{SiO}_2$  and KCl, the number concentration decreased slightly, and the modal diameter of the nucleation mode decreased to 15 nm. When using a dilution ratio



**Figure 2.** The model-aerosol-generating system.  $\text{SiO}_2$  particles were generated in the first reactor and KCl vapor was generated in the second reactor.



**Figure 3.** (a) Particle number size distribution of the laboratory generated aerosol, using different probe dilution ratios. (b) Particle mass size distribution of laboratory-generated aerosol, using different precursor compositions and dilution ratios.

of 1:100 the nucleation mode vanished almost completely. The size distribution above 70 nm was not affected by adding KCl or by using different dilution ratios. The mass size distributions calculated from the SMPS data are given in Figure 3b. An effective particle density of  $2 \text{ g/cm}^3$  was assumed. When using the 1:10 dilution ratio and feeding KCl vapor only, a distinct nucleation mode around 35 nm was generated. When adding the SiO<sub>2</sub> particles, an additional mode around 220 nm was generated. When closing down the KCl feed, the nucleation mode disappeared, while the 220 nm mode decreased slightly. When repeating the same procedure, using a dilution ratio of 1:33, the KCl-induced nucleation mode decreased by a factor of 10. When increasing the dilution ratio to 1:100, the nucleation mode was further decreased by a factor of about 1000. These results indicate that raising the dilution ratio will effectively prevent KCl vapor from affecting the original SiO<sub>2</sub> particle distribution. The total mass of the SiO<sub>2</sub> mode decreased slightly when the KCl feed was closed, but the particle size did not increase. This indicates that no significant condensation on the existing particles occurred. The slight decrease in concentration is prob-

ably an effect of fewer losses in the reactor and in the sampling system when KCl vapor was present, e.g., by neutralization of the highly charged SiO<sub>2</sub> particles by ionized KCl in the hot reactor.

There are several possible mechanisms, which may explain the enhanced separation of KCl vapor at high dilution ratios, including:

1. Increased ratio of wall condensation to particle formation.
2. Formation of smaller KCl nuclei, which are exceedingly lost by deposition onto the probe walls and tubing, by means of diffusion.
3. A nucleation mode below 10 nm is formed, which is not detectable by the instruments used.

The separation effect might be a result from a combination of the listed mechanisms. The first mechanism allows for a more complete and controlled separation process, and therefore the subsequent discussion will be focused on this. The last two mechanisms are rather straightforward and will not be discussed further.

Upstream from the cooling section, directly after dilution, KCl will be present as a vapor due to the high temperature of the dilution gas. When the aerosol is transported down the probe it will be cooled by the walls. As the gas temperature drops below 400°C, the vapor pressure of KCl is close to zero and the KCl will be present as solid phase material. There are two competing routes for the condensing material. Firstly, the vapors might condense on the internal surfaces of the probe. Secondly, the gaseous KCl might undergo gas-to-particle formation by nucleation, coagulation, and particle surface condensation, thereby contributing to the total aerosol particle mass. Gas-to-particle conversion by condensation will occur only in volumes where the KCl saturation ratio (the ratio of partial pressure to vapor pressure) exceeds unity, and homogenous nucleation is expected to place at nucleation fronts where the saturation ratio is much higher than unity. From the results given in Figure 3b, it is clear that when using low dilution ratios, homogenous nucleation is the dominating gas-to-particle route, and that only a small fraction of the KCl will deposit on the SiO<sub>2</sub> particles. According to the classical nucleation theory, the rate of particle formation  $J$  (m<sup>-3</sup> s<sup>-1</sup>) is given by (Seinfeld and Pandis 1998)

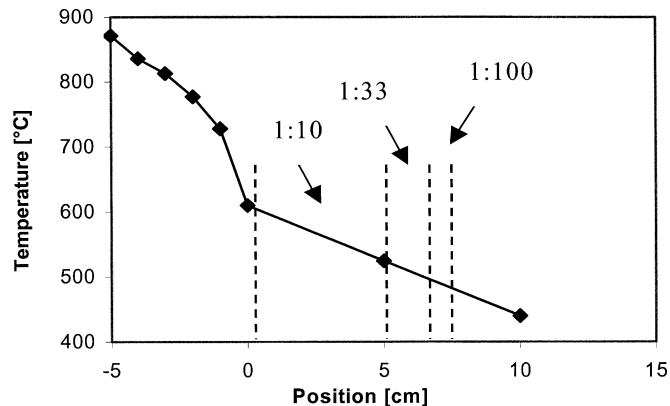
$$J = \left( \frac{2\sigma}{\pi m_1} \right)^{1/2} \frac{v_1 N_1^2}{S} \exp \left( - \frac{16\pi v_1^2 \sigma^3}{3(k_B T)^3 (\ln S)^2} \right) \quad [1]$$

where  $\sigma$  (J m<sup>-2</sup>) is the surface tension;  $m_1$  (kg) and  $v_1$  (m<sup>3</sup>) are the molecular mass and volume, respectively;  $N_1$  (m<sup>-3</sup>) is the concentration of monomers;  $S$  is the saturation ratio;  $k_B$  (J K<sup>-1</sup>) is Boltzmann's constant; and  $T$  (K) is the gas temperature.

When the dilution ratio is increased, the KCl vapor concentration, and consequently also the saturation ratio, will be decreased at any specific position upstream from the nucleation front. As indicated by Equation (1), the nucleation rate, and therefore also the position of the nucleation front, has a strong dependence on the saturation ratio. Consequently, the nucleation front will be displaced and nucleation will take place at lower temperatures as the dilution ratio is raised. This will allow for a larger fraction of the vapor to condense onto the walls of the probe, diminishing the fraction available for particle formation.

In order to estimate the position of the nucleation front, the axial temperature profile in the inlet tube and the cooling tube was measured using a thermocouple inserted in the probe. The axial temperature profile along the dilution probe center is given in Figure 4. Varying the dilution ratio had only minor effects on the temperature profile, since most of the heat transport in and out of the dilution probe was through the probe walls.

The zero position in the figure corresponds to the entrance of the cooling tube and the -5 cm position to the probe inlet. The temperature measurements are approximate because the thermocouple is exposed to radiation exchange with the surrounding surfaces. For simplicity, no radial temperature gradient is assumed and the tube wall is regarded as a concentration sink. In the laboratory experiments, when adding 20 mg/m<sup>3</sup> of KCl in the reactor it might be assumed that a nucleation front in the



**Figure 4.** Axial temperature distribution in the dilution probe. The given position is from the tip of the cooling section. The positions between the dashed lines correspond to positions where the saturation ratio of 1–30 might be expected.

dilution probe will appear at a position in the cooling section where the saturation ratio is between 1 and 30. This was calculated from thermodynamic data to correspond to a position 0–5 cm in the cooling section, when a dilution ratio of 1:10 was used, and assuming that all KCl vapor is transported to the nucleation front. For dilution ratios of 1:33 and 1:100, the nucleation front would appear at slightly lower temperatures in the 5–7 cm and 7–8 cm sections, respectively. The assumed positions of the nucleation fronts are very close, especially for the two highest dilution ratios used. Now, all KCl vapor will not be transported to the nucleation front, but a fraction will condense on the cooled probe walls. Assuming that the walls are perfect sinks, the fraction of KCl vapor penetrating to the nucleation front may be approximated by the expression (Willeke et al. 1993)

$$\eta_{diff} = \exp \left( - \frac{\pi d L V_{diff}}{Q} \right) \quad [2]$$

where  $d$  is the diameter of the tube,  $L$  is the nucleation front position,  $V_{diff}$  is the deposition velocity, and  $Q$  is the volumetric flow. Since the estimated positions of nucleation fronts for the dilution ratios used are very close, the fraction of material available for nucleation should be similar, at least for the two highest dilution ratios used. This is opposed by the experimental results, where it seems that nucleation is actually completely suppressed when a dilution ratio of 1:100 is used, and there must be some other mechanism to account for this effect. According to Equation (1), the nucleation rate in the probe is thermodynamically determined mainly by the saturation ratio in the exponential term. However, since the initial vapor concentration is very low, and the cooling rate is high, the nucleation might be determined not only thermodynamically but also by the collision frequency of the monomers and clusters formed. In the collision-controlled regime, the nucleation rate and consequently also the nucleation front position will be more linearly dependent on concentration.

Therefore, the fraction of material available for particle formation at the nucleation front will be inversely dependent on the exponential of the inlet concentration according Equation (2). This is more in line with the experimental results obtained.

There are probably several options of optimizing the sampling method in order to control the deposition and nucleation process better. These options include changing the probe design and flow conditions. They also include optimizing axial and radial temperature distributions. Probe design, flow conditions, and rates of mass-heat transport are complexly and strongly coupled properties, and any optimization will be facilitated by a more thorough analysis of the transport phenomena and aerosol dynamics involved. The system also needs to be calibrated for particle-size-dependent losses in order to establish a truly quantitative method. However, optimization and calibration is not within the scope of this investigation.

When sampling hot aerosols with high humidity, or gases containing volatile and semivolatile compounds, dilution is often used in order to avoid condensation. This is accomplished by decreasing the partial pressure of the vapors to assure that the saturation ratio is kept below unity, also after cooling the gas. It should be stressed that this mechanism is not at hand for avoiding condensation of most fly ash precursors because the vapor pressure of, e.g., KCl, is close to zero below 400°C.

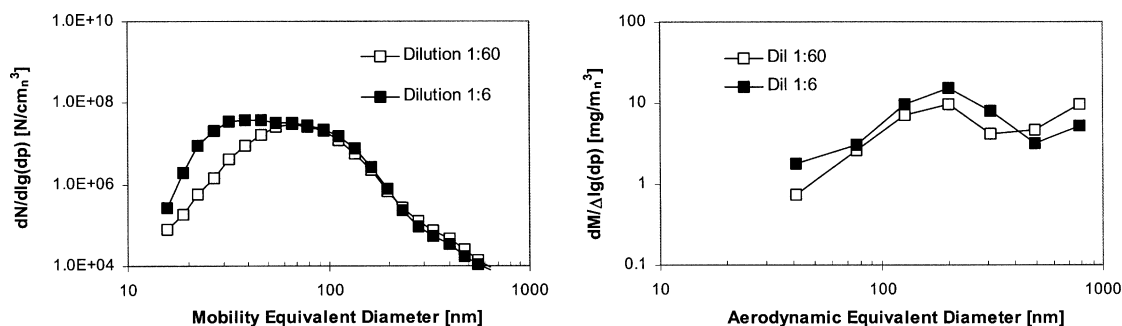
### Results from Field Measurements

The flue gas was sampled at 780°C in the CFB boiler using low and high dilution ratios of 1:6 and of 1:60, respectively. The number and mass size distributions using the two different dilution ratios (corrected for dilution) are shown in Figure 5. The high dilution ratios (1:60) needed for suppression of the nucleation mode in the CFB boiler was slightly lower than what was used in the laboratory experiments (1:100). This was probably an effect of a lower inorganic vapor concentration in the boiler than what was used in the experimental setup. In fact, the main problem during the field measurements was not to suppress the nucleation mode but to produce it by using a sufficiently low dilution ratio. Probably a more distinct nucleation mode would have appeared if a dilution ratio lower than 1:6 could have been

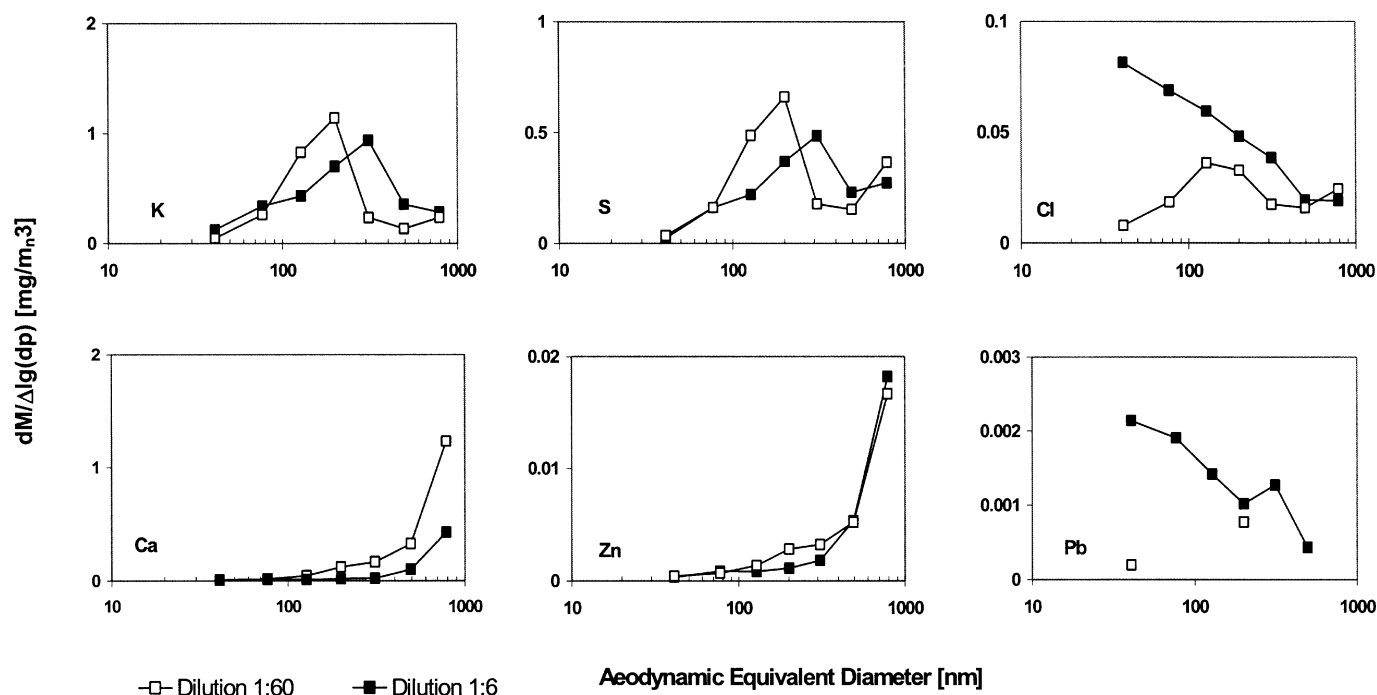
used. However, a lower dilution ratio was prevented by the high water content of the flue gas, because lower dilution ratios would have caused condensation of water in the sampling line.

One alternative to produce the nucleation mode is to use a porous dilution probe for the low dilution sampling. In the porous dilution probe the gas is cooled and diluted through the walls of a sintered tube, and the losses to the walls are minimized. A setup with parallel sampling lines including a high-dilution probe and a porous dilution probe will probably facilitate the investigation of the particle-vapor distribution, especially the quantitative analysis.

The number concentration for the high dilution sample was  $1.2 \times 10^7$  particles/cm<sup>3</sup>, the GMD was 68 nm, and the GSD was 1.47. When using the low dilution, a nucleation mode around 40 nm was generated and the number concentration was  $2.2 \times 10^7$  particles/cm<sup>3</sup>. The GMD was 52 nm and the GSD was 1.61. The mass size distribution given by the gravimetric analysis of the DLPI substrates are not highly accurate due to the small mass of the particles deposited on the substrates, especially during the high dilution sampling. Nevertheless, the DLPI results are consistent with the SMPS results, and the nucleation mode is exposed as an enhanced mass on the first impactor stage ( $d_{ae} = 40$  nm) when using the low dilution ratio. The total fine particle mass concentration ( $d_{ae} < 1000$  nm) was 7.7 mg/m<sup>3</sup> when using the high dilution ratio, and 9.2 mg/m<sup>3</sup> when using the low dilution ratio. The total particle mass was also calculated from the SMPS data, using an effective particle density of 2 g/cm<sup>3</sup>. The calculated mass concentration was 8.8 mg/m<sup>3</sup> and 10.3 mg/m<sup>3</sup> for the low and high dilution ratios, respectively. The nucleation mode appearing during the low dilution sampling is most probably formed from vapors nucleating in the dilution probe. When the high dilution ratio is used, the main part of the vapors will deposit on the internal walls of the probe, and the result is more correctly representing the actual particle size distribution in the hot flue gas inside the boiler. The concentration of the condensing material seems to be much lower than what would be expected for wood-based fuels at 780°C, i.e., alkali chlorides present as vapors. This is most likely an effect of the high sulphur content in the fuel, which will favor the formation of alkali



**Figure 5.** Particle number size distributions and mass size distributions of flue gas from the CFB boiler, using two different dilution ratios.



**Figure 6.** Elemental mass size distributions of the flue gas from the CFB boiler, using two different dilution ratios.

sulphates that have a much lower vapor pressure at 780°C than the corresponding chlorides.

The elemental mass size distribution as given by PIXE analysis of the impactor substrates is shown in Figure 6. When using the high dilution ratio (1:60), the elemental size distributions can be sorted into two types. The first one is represented by the distribution of K, S, Cl, and possibly Pb, where the mass is concentrated in a single mode around 200 nm. This type of elemental size distribution indicates that the elements contribute to the particle phase by gas-to-particle conversion inside the boiler, upstream from the probe position. The unimodal distribution, i.e., the absence of any nucleation mode, also indicates that there was no nucleation of these elements close to sampling position. The second distribution type is represented by the distribution of Ca and Zn, where the elemental mass is increasing monotonously with particle size in the submicron size range, indicating a single mode in the supermicron range. This type of size distribution indicates that the elements are mainly associated with a supermicron fraction of refractory components that were not vaporized during the combustion process, but transported through the boiler as fly ash particles.

When changing to the low dilution ratio (1:6), the elemental size distribution for K, S, Ca, and Zn were unaltered, showing that the major fraction of these elements were present as particles at the sampling position. In contrast, the Cl and Pb size distributions changed dramatically, indicating a nucleation mode that was formed from vapors nucleating inside the probe. The results suggest that more than half of the Cl and almost all Pb was

present as inorganic vapors at the sampling position. According equilibrium calculations made using a commercial software (HSC Chemistry, Outokumpu Research Oy, Pori, Finland), Cl should be present as alkali chloride vapors or HCl gas at 780°C, and the relatively high fraction of Cl in the particle phase was not anticipated. Further investigations are needed in order to establish if Cl might be associated with some other elements, e.g., calcium, which might form submicron particles by gas-to-particle conversion at higher temperatures.

## CONCLUSIONS

The objective of this study was to design and experimentally examine a sampling method for high-temperature aerosols from biomass combustion in which nucleation and condensation from gas-to-particle phase is avoided. The sampling method included a high-temperature dilution probe in which the sampled gas is diluted and then cooled. Experimental results from sampling of an aerosol with a known concentration of SiO<sub>2</sub> particles and KCl vapor, showed that when using a high dilution ratio the KCl vapor was effectively separated from the aerosol by condensation onto the cooled probe walls. When a lower dilution ratio was used, the KCl vapor generated a nucleation mode. The sampling method was also used in a CFB boiler fired with forest residues. The results indicated that the major fraction of Ca, K, S, and Zn were present as particles at 780°C, whereas substantial fractions of Cl and Pb were present as gases. These results are in line with previous experimental and theoretical results, and indicate that



the method can be used for efficient and precise characterizing of high-temperature combustion aerosols containing inorganic vapors. The method allows for measurements of transients using online instruments as well as for extracting time-averaged samples using filters and impactors.

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